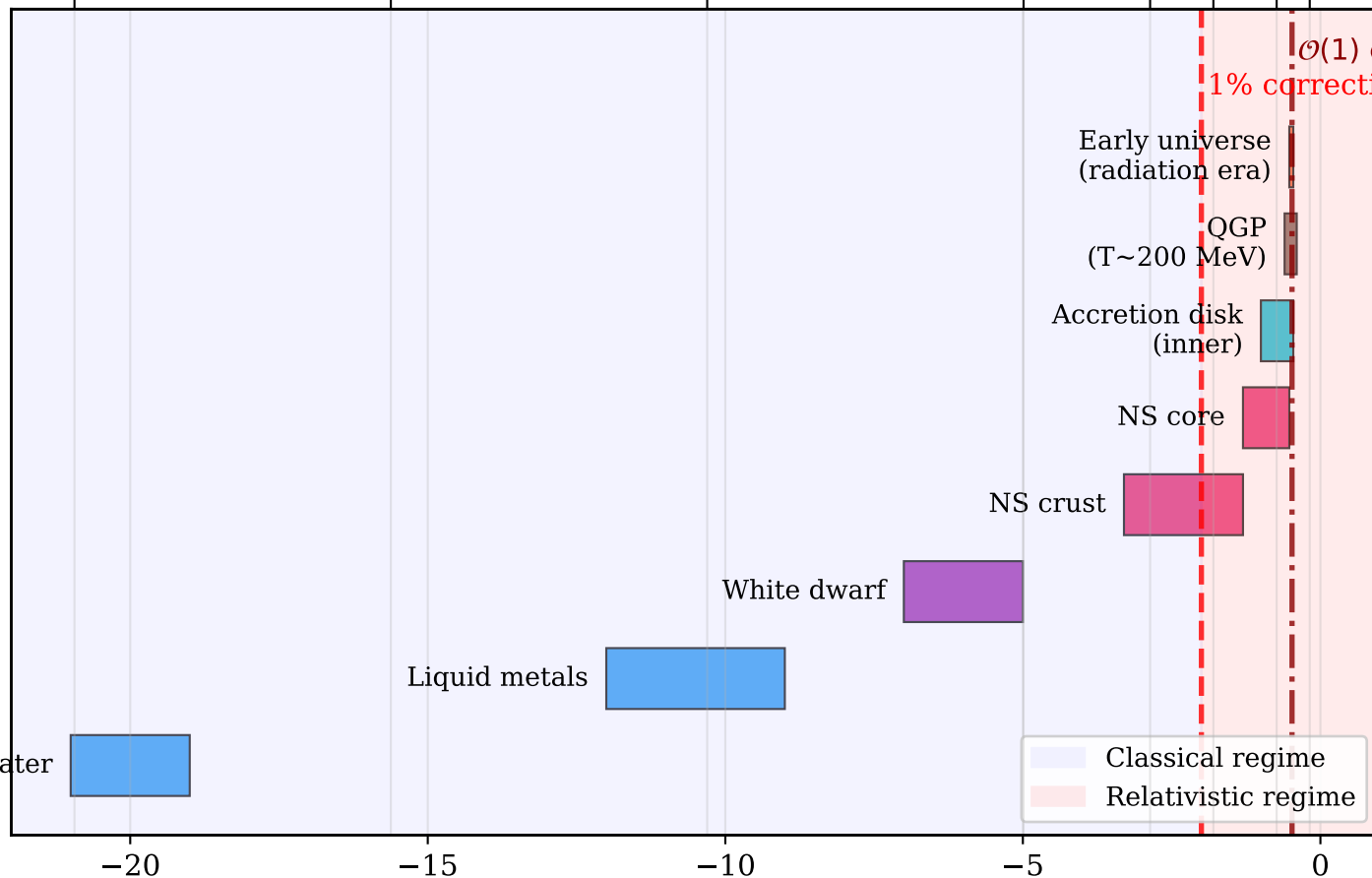


Relativistic Compactness Parameter ξ Across Astrophysical Regimes

Relativistic correction to Ra_c : $\Delta Ra/Ra \approx \xi$

1e-18% 1e-13% 1e-08% 1e-03% 0.1% 1.0% 10% 33%



Classical regime
Relativistic regime

$\mathcal{O}(1)$ correction ($\xi = 1/3$)
1% correction ($\xi = 0.01$)

$\log_{10} \xi$ where $\xi = p_0/(\epsilon_0 c^2)$